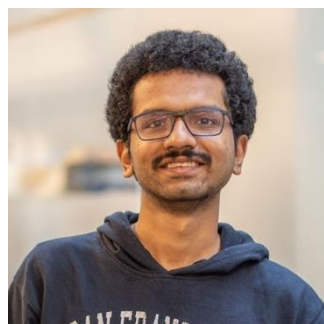


Min CSPs on Complete Instances II:

Polylogarithmic Approximation for Min-NAE-3-SAT

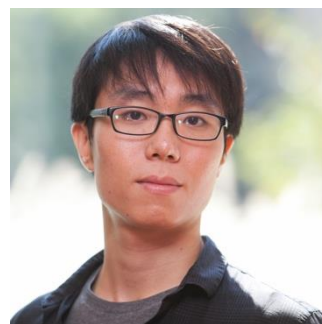
by

Aditya Anand



UMich

Euiwoong Lee



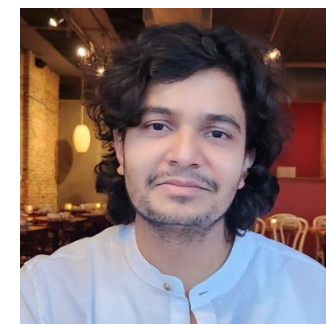
UMich

Davide Mazzali



EPFL

Amatya Sharma



UMich

A Puzzle

- 3-SAT
 - Check in poly time: Satisfiable or not?
 - No, NP-Hard in general.
 - What if there is a clause for every 3 variables? **“Complete” instance.**

“Decision”
3-SAT



r/mathmemes • 2 mo. ago
SlipPuzzleheaded7009



Guys, I Did It: My Proof That $P = NP$. Fingers Crossed for the Fields Medal



A Simple Resolution of $P = NP$

u/SlipPuzzleheaded7009

Abstract

A Puzzle

- 3-SAT

- Check in poly time: Satisfiable or not?
 - No, NP-Hard in general.
 - What if there is a clause for every 3 variables? **“Complete” instance.**

“Decision”
3-SAT

e.g. **Complete** 3-SAT:

$(x_1 \vee \neg x_2 \vee x_3),$
 $(x_1 \vee \neg x_2 \vee \neg x_4),$
 $(x_1 \vee x_3 \vee x_4),$
 $(x_2 \vee \neg x_3 \vee x_4).$

Thm. For **complete** 3-SAT on n variables, $|(\text{satisfying assignment})| = O(n^2).$

- Algorithm:

- Order variables $1, \dots, n$
- $A_3 \leftarrow$ List of all satisfying assignments to first three variables (one clause)
- For $i = 4$ to n :
 - Consider clauses only on first i variables. **Complete Instance on i variables**
 - $A_i \leftarrow$ Extend each assignment of A_{i-1} by trying all values of i^{th} variable, that satisfy all the $\binom{i}{3}$ clauses

Simple
VC-Dim
Argument

Thm. Can solve **complete** 3-SAT in $\text{poly}(n)$ time.

[ALMS25]

A Puzzle

- 3-SAT

- Check in poly time: Satisfiable or not?
 - No, NP-Hard in general.
 - What if there is a clause for every 3 variables? **“Complete” instance.**

“Decision”
3-SAT

e.g. **Complete** 3-SAT:

$(x_1 \vee \neg x_2 \vee x_3),$
 $(x_1 \vee \neg x_2 \vee \neg x_4),$
 $(x_1 \vee x_3 \vee x_4),$
 $(x_2 \vee \neg x_3 \vee x_4).$

Thm. For **complete** k -SAT/CSP on n variables, $|(\text{satisfying assignment})| = O(n^{k-1})$.

- Algorithm:

- Order variables $1, \dots, n$
- $A_3 \leftarrow$ List of all satisfying assignments to first three variables (one clause)
- For $i = 4$ to n :
 - Consider clauses only on first i variables. **Complete Instance on i variables**
 - $A_i \leftarrow$ Extend each assignment of A_{i-1} by trying all values of i^{th} variable, that satisfy all the $\binom{i}{3}$ clauses

Simple
VC-Dim
Argument

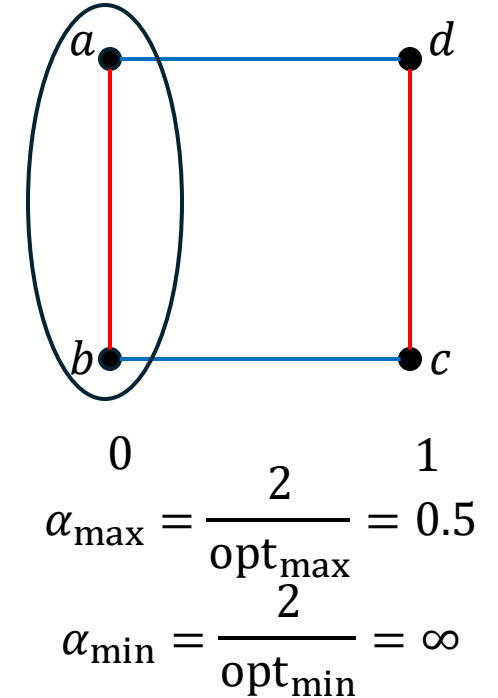
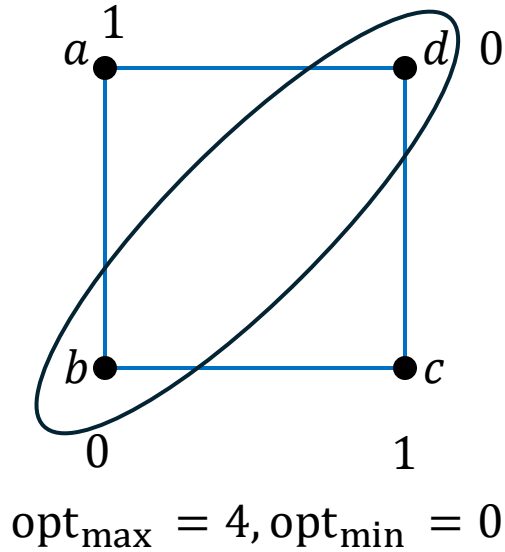
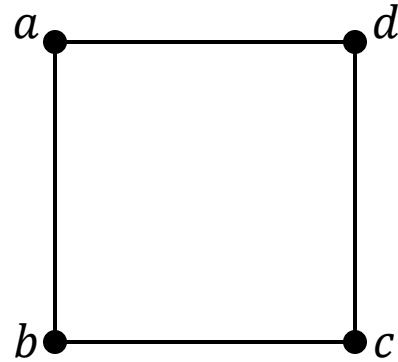
Thm. Can solve **complete** k -SAT/CSP in $\text{poly}(n)$ time. [ALMS25]

CSP example

(Not-All-Equal) NAE-3-SAT:

- n Boolean variables $V = \{x_1, x_2, \dots, x_n\}$
- m Clauses:
 - $(x_1, \neg x_3, x_4), (\neg x_2, \neg x_3, x_4), \dots$
 - Predicate: Clause is **unsatisfied** iff literals are all same
- Objective: Find an assignment that...
 - Satisfy all the clauses NAE-3-SAT
 - (Approximately) Maximize the #satisfied clauses **Max**-NAE-3-SAT
 - (Approximately) Minimize the #unsatisfied clauses **Min**-NAE-3-SAT

e.g. **Max**-CUT vs **Min**-(Un)CUT



- Objective: Find an assignment that...
 - Satisfy all the clauses NAE-3-SAT
 - (Approximately) Maximize the #satisfied clauses Max-NAE-3-SAT
 - (Approximately) Minimize the #unsatisfied clauses Min-NAE-3-SAT

$\text{opt}_{\min} = 0 \Leftrightarrow \text{alg}_{\min} = 0$
Any finite approximation
decides satisfiability

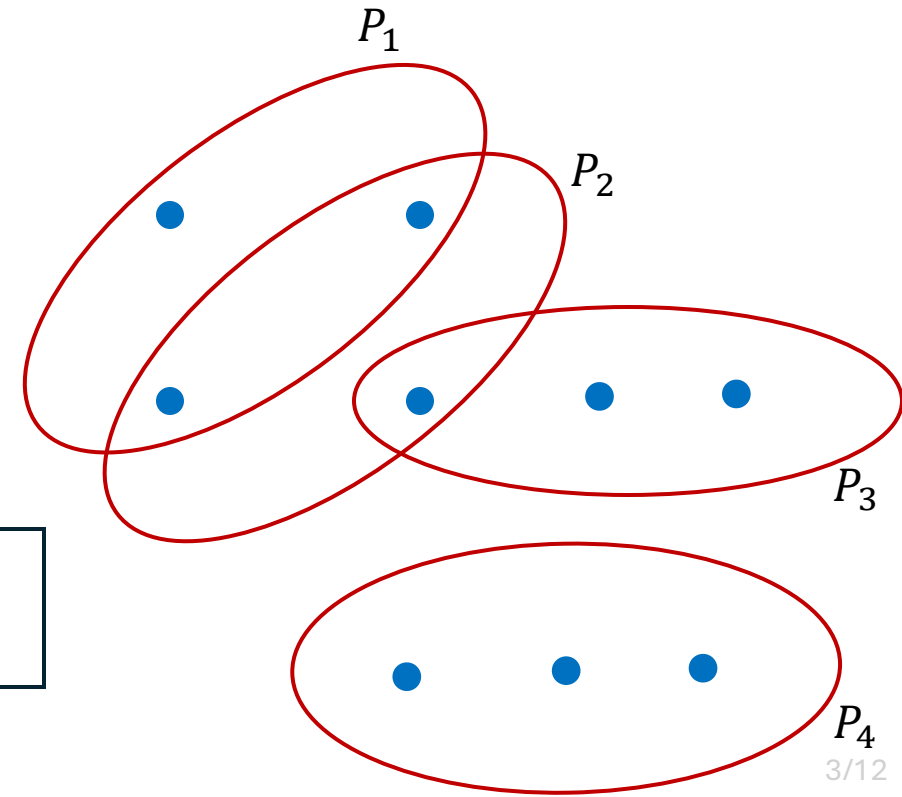
Constraint Satisfiability Problems

e.g. NAE: $(x_1, \neg x_3, x_4)$
CSP: $c = (x_1, x_3, x_4)$
Predicate $P_c(0,1,0), (1,0,1) \mapsto \text{unsat}$
 $(0,0,0) \dots \mapsto \text{sat}$

k -CSP:

- n Boolean Variables
- m Clauses with
 - k -variables each clause
 - **Predicate** $P_j: \{0,1\}^k \mapsto \{\text{sat}, \text{unsat}\}$ for each clause j
- Objectives: k -CSP, Max- k -CSP, Min- k -CSP
- Instance is a k -uniform hypergraph with Predicates
- Structured Instances:
 - Dense ($\Omega(n^k)$ edges),
 - Expanding,
 - **Complete ($\binom{n}{k}$ edges).**

There's a clause for every k variables.



History: Max-CSPs

General Instances

- Tools
 - Linear, Semidefinite Programming
 - Probabilistically Checkable Proofs
 - Unique Games Conjecture
- **Optimal* Approximation** Algorithms
 - Max-3-SAT, Max-3-LIN, Max-CUT, Unique Games [Has01, Kho02, KKMO07]

Structured Instances (dense/complete/expanding)

- Tools
 - Random Sampling
 - Convex Hierarchies
 - Regularity Lemmas
- **Every Max-CSP** has a **PTAS** on **dense/complete/expanding** instances via **any tool**
[AKK95, BFdLVK03, ADLVKK03, dLVKKV05, MS08, KS09, BHHS11, Yar14, MM15, FLP16, dLVKM07, AKK+08, BRS11, GS11, YZ14, AJT19, JQST20, BBK+21, FK96, COCF10, OGT13, JST21]

History: Min-CSPs

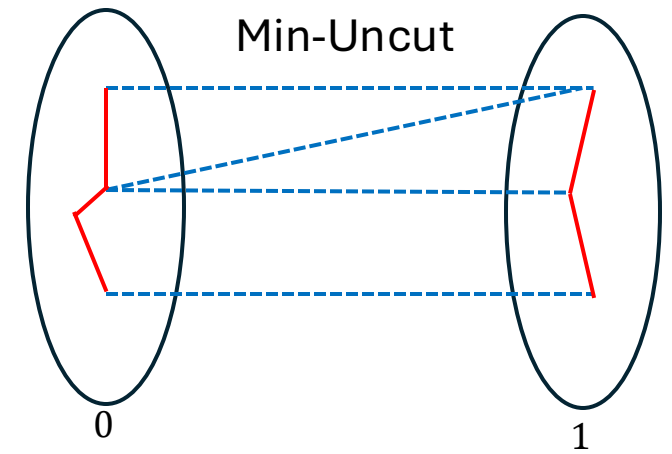
Goal: Minimize #**unsatisfied** clauses
e.g. Min-Uncut, Min-2-SAT, Min-3-SAT

General Instances

- Unlike Max-CSPs, not all Min-CSPs admit $O(1)$ -approx.
- Structural Characterization identifies **optimal approximations** as one of $1, O(1), \text{poly log}(n), \text{poly}(n), \infty$ [KSTW01]

Structured Instances (dense/expanding/complete)

- Min Uncut & Unique Games
 - $O(1)$ -approximation for on dense/complete/expanding instances [BFdLVK03, KS09, GS11, MdMMN23]
- Fragile CSPs
 - PTAS on dense/complete instances [KS09]
- Min-2-SAT (aka Min-2CNF Deletion)
 - UGC-Hard to get any constant approximation on dense instances. [CC04]
 - Best known $O(\sqrt{\log n})$ on dense/complete instances [ACMM05]



Min-CSPs on Complete Instances

- Complete Min-2-SAT admits an $O(1)$ -approximation in $\text{poly}(n)$ -time. [ALS25]
 - Separates Dense ($\Omega(1)$) vs Complete ($O(1)$).
- What next? NAE-3-SAT?
 - Max-NAE-3-SAT: ≈ 0.9089 (tight) approximation (general), PTAS (dense). [BHPZ21]
 - Min-NAE-3-SAT: ∞ as it is NP-Hard to decide (general/dense).

Thm. $\text{poly}(n)$ time algorithm for k -CSP on complete instances

$\Rightarrow$ $\text{poly}(n)$ time algorithm to decide complete NAE-3-SAT

Thm. $n^{\text{poly}(\log n)}$ time $O(\log^6(n))$ -approx for Min-NAE-3-SAT on complete instances

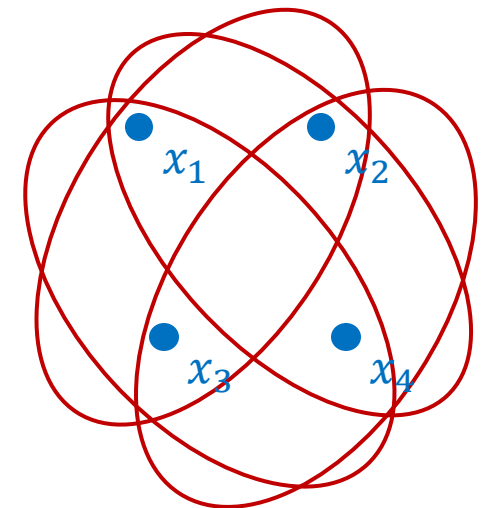
Min-NAE-3-SAT on Complete Instance

Min-NAE-3-SAT:

- Clause is **unsatisfied** iff literals are all same
- Objective is to Minimize #unsatisfied clauses
- **Complete**: A clause for every 3-tuple of variables

e.g. complete Min-NAE-3-SAT:

$(x_1, \neg x_2, x_3),$
 $(x_1, \neg x_2, \neg x_4),$
 $(x_1, x_3, x_4),$
 $(x_2, \neg x_3, x_4).$



Sherali-Adams LP & Conditioning

- Solve the r -round SA LP:

$$\text{LP} := \min \sum_{\text{all clauses } C} E_{\hat{\mu}} [\text{val}(C)]$$

$\hat{\mu} \in \text{All } r\text{-local distributions over assignments to } n \text{ variables.}$

- For all $S \subseteq V, |S| \leq r$ and $\alpha \in \{0,1\}^S$, $\text{var } x_{S,\alpha} = \widehat{\Pr}[S \leftarrow \alpha]$ in OPT. Local Consistency Constraints.
- $\text{val}(C) = 1$ iff clause is unsat. $E_{\hat{\mu}} [\text{val}(C)] = \sum_{\alpha \text{ s.t. } \text{val}(C)=1} x_{C,\alpha}$.
- Can we round it?
 - Conditioning (Dense Max- k -CSPs):
 - Sample $T \subseteq V, |T| \leq r - 1$
 - Sample $\alpha \sim \{0,1\}^T$ from the local distribution $\hat{\mu}$
 - Obtain the conditional distribution: $\mu' = \hat{\mu}|T \leftarrow \alpha$
 - For all $i \in V \setminus T$, sample i 's value from the conditional distribution μ' (using local distribution of $T \cup \{i\}$)
 - **Variance Reduction** [BRS11, GS11, RT12, YZ14]
 - Condition on r variables, the correlation among every t variables, is at most $O(2^t/\sqrt{r})$.

Sherali-Adams LP & Conditioning

- Solve the r -round SA LP:

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 - **Variance Reduction** [BRS11, GS11, RT12, YZ14]
 - Condition on $1/\epsilon^2$ variables, the correlation among every k variables, is at most $O(2^k \epsilon)$.

Sherali-Adams LP & Conditioning

- Solve the r -round SA LP:

$$\text{LP} := \min \sum_{\text{all clauses } C} E_{\hat{\mu}} [\text{val}(C)]$$

$\hat{\mu} \in \text{All } r\text{-local distributions over assignments to } n \text{ variables.}$

- For all $S \subseteq V$, $|S| \leq r$ and $\alpha \in \{0,1\}^S$, $\text{var } x_{S,\alpha} = \widehat{\Pr}[S \leftarrow \alpha]$ in OPT. Local Consistency Constraints.
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 - **Variance Reduction** [BRS11, GS11, RT12, YZ14]
 - Condition on $1/\epsilon^2$ variables, the correlation among every k variables, is at most $O(\epsilon)$.
 - Round $\approx_{\epsilon n^k} \text{opt}_{\max}$, where $\epsilon = 1/\sqrt{r}$, can assume $\text{opt}_{\max} = \Omega(n^k)$
 - This is bad for Min-NAE-3-SAT. If $\text{opt} = 0$, we can't afford any additive error of ϵn^3 .

Conditioning + Thresholding

- How about an additive error of $O(LP)$?
 - $LP = \delta n^3$
 - If $\delta \geq 1/\log^{100} n$, any assignment is $O(1/\delta)$ -approximation
 - Let's only round "almost-fixed" variables.
- Variable x is **almost fixed** w.r.t. a distribution μ if
 - $\Pr_\mu[x = F] \leq \tilde{O}(\delta)$, or $\geq 1 - \tilde{O}(\delta)$

x	y	z
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Conditioning + Thresholding

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		x	y	z
$\tilde{O}(\delta) \geq$	{	1	0	0
		1	0	1
		1	1	0
		1	1	1
$1 - \tilde{O}(\delta) \leq$	{	1	0	0
		1	0	1
		1	1	0
		1	1	1

Conditioning + Thresholding

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$$LP_{\text{new}} = LP + \tilde{O}(\delta) n^3$$

	x	y	z
$\tilde{O}(\delta) \geq \left\{ \right.$	1	0	0
	1	0	1
	1	1	0
	1	1	1
$1 - \tilde{O}(\delta) \leq \left\{ \right.$	1	0	0
	1	0	1
	1	1	0
	1	1	1

Conditioning + Thresholding

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 - $LP = \delta n^3$
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- Variable x is **almost fixed** w.r.t. a distribution μ if
 - $\Pr_\mu[x = F] \leq \tilde{O}(\delta)$, or $\geq 1 - \tilde{O}(\delta)$
- Conditioning + Thresholding:
 - Sample a pair $(x, z) \sim \binom{V}{2}$
 - Sample α_x, α_z from the local distribution $\hat{\mu}_{(x,z)}$
 - Obtain the conditional distribution: $\mu' = \hat{\mu}|((x, z) \leftarrow (\alpha_x, \alpha_z))$
 - Fix all almost fixed $y \in V \setminus T$ (w.r.t. μ') to the **ruling value**
 - Recurse

$$LP_{\text{new}} = LP + \tilde{O}(LP)$$

$\tilde{O}(\delta) \geq \left\{ \begin{array}{c} \text{Row 4} \\ \text{Row 5} \\ \text{Row 6} \\ \text{Row 7} \end{array} \right.$
 $1 - \tilde{O}(\delta) \leq \left\{ \begin{array}{c} \text{Row 2} \\ \text{Row 3} \\ \text{Row 8} \end{array} \right.$

x	y	z
1	0	0
1	0	1
1	1	0
1	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Thresholding

Algorithm

- Conditioning + Thresholding:

- Sample and condition on $X = \text{poly}(\log n)$ variables and assignment $\alpha_X \sim \hat{\mu}_X$.
- Obtain the conditional distribution $\mu' := \hat{\mu}|(X \leftarrow \alpha_X)$
- Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
- Recurse

Assuming we fix $\Omega(n)$ variables every time:

$$\text{Time: } \left[2^{\text{poly}(\log n)} \cdot \binom{n}{\text{poly}(\log n)} \right]^{O(\log n)} = n^{\text{poly}(\log n)} \rightarrow \boxed{\approx n^{|X|}}$$

Analysis

- “**Double**” Conditioning + Thresholding:

- Sample and condition on $|R| = r = \text{poly}(\log n)$ variables and assignment $\alpha_R \sim \hat{\mu}_R$.
- Sample and condition on $|T| = t$ variables and assignment $\alpha_T \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
- Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
- Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
- Recurse

Condition
on $X = R \cup T$

- In each iteration i , there is a **good choice**:

- Almost fixes $\Omega(n)$ variables

(bound the runtime by $n^{\text{poly}(\log n)}$)

- p_{fix}

- Bound the increase in obj

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

- p_{obj}

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Analysis

- “**Double**” Conditioning + Thresholding:

- Sample and condition on $|R| = r = \text{poly}(\log n)$ variables and assignment $\alpha_R \sim \hat{\mu}_R$.
- Sample and condition on $|T| = t$ variables and assignment $\alpha_T \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
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- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables

(bound the runtime by $n^{\text{poly}(\log n)}$)

- p_{fix}

- Bound the deviation of LP_i :

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

- $p_{obj} \geq 1/\log n$

- Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$

- $LP_{O(\log n)} \leq \text{poly}(\log n) \cdot LP_0$

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Analysis

- “**Double**” Conditioning + Thresholding:

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- Sample and condition on $|T| = t$ variables and assignment $\alpha_T \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
- Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
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- p_{fix}

- Bound the deviation of LP_i :

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

- $p_{obj} \geq 1/\log n$

- Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$

- $ALG \leq \text{poly}(\log n) \cdot \text{opt}$

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Analysis

- “**Double**” Conditioning + Thresholding:

- Sample and condition on $|R| = r = \text{poly}(\log n)$ variables and assignment $\alpha_R \sim \hat{\mu}_R$.
- Sample and condition on $|T| = t = 2$ variables and assignment $\alpha_T \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
- Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
- Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
- Recurse

Condition
on $X = R \cup T$

- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables (bound the runtime by $n^{\text{poly}(\log n)}$)
 - $p_{fix} \geq 1/1000$ — randomly sample $t = 2$ variables
- Bound the deviation of LP_i : (bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)
 - $p_{obj} \geq 1/\log n$
 - Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$
 - $ALG \leq \text{poly}(\log n) \cdot \text{opt}$

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Analysis

- “**Double**” Conditioning + Thresholding:

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- Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
- Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
- Recurse

Condition
on $X = R \cup T$

- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables

(bound the runtime by $n^{\text{poly}(\log n)}$)

- $p_{fix} \geq 1/1000$

randomly sample $t = 2$ variables

- Bound the deviation of LP_i :

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

- $p_{obj} \geq 1/\log n$

- Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$

- $ALG \leq \text{poly}(\log n) \cdot \text{opt}$

$$(1 - p_{fix}) + (1 - p_{obj}) \not\leq 1$$

Analysis

- “**Double**” Conditioning + Thresholding:

- Sample and condition on $|R| = r = \text{poly}(\log n)$ variables and assignment $\alpha_R \sim \hat{\mu}_R$.
 - Sample and condition on $|T| = t = 100 \log \log n$ variables and assignment $\alpha_R \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
 - Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
 - Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
 - Recurse
- Condition
on $X = R \cup T$

- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables

(bound the runtime by $n^{\text{poly}(\log n)}$)

 - $p_{\text{fix}} \geq 1 - (1 - 1/1000)^{100 \log \log n}$ — randomly **independently** sample $t = 100 \log \log n$ vars
- Bound the deviation of LP_i :

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

 - $p_{\text{obj}} \geq 1/\log n$
 - Conditioning: $E_{X \leftarrow \alpha_X}[\text{LP}_i] = \text{LP}_{i-1}$
 - $\text{ALG} \leq \text{poly}(\log n) \cdot \text{opt}$

$(1 - p_{\text{fix}}) + (1 - p_{\text{obj}}) < 1$

Analysis

- “**Double**” Conditioning + Thresholding:

- Sample and condition on $|R| = r = \text{poly}(\log n)$ variables and assignment $\alpha_R \sim \hat{\mu}_R$.
- Sample and condition on $|T| = t = 100 \log \log n$ variables and assignment $\alpha_R \sim \hat{\mu}_T | R \leftarrow \alpha_R$.
- Obtain the conditional distribution $\mu' := \hat{\mu} | (R \leftarrow \alpha_R, T \leftarrow \alpha_T)$
- Fix all almost fixed $y \in V \setminus T$ (w.r.t μ') to their ruling assignment.
- Recurse

Condition
on $X = R \cup T$

- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables

- $p_{fix} \geq 1 - 1/\log^{50} n$

(bound the runtime by $n^{\text{poly}(\log n)}$)

- Bound the deviation of LP_i :

- $p_{obj} \geq 1/\log n$

- Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$

- $ALG \leq \text{poly}(\log n) \cdot \text{opt}$

randomly **independently** sample $t = 100 \log \log n$ vars

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Analysis

Variance Reduction [BRS11, GS11, RT12, YZ14]

- Condition on r variables, the correlation among every t variables, is at most $O(2^t/\sqrt{r})$
- “First” conditioning **condition on $r = \text{poly}(\log n)$ variables**,
 - For every $t = 100 \log \log n$ variables:

Sampling t variables **independently** $\approx_{O(1)}$ Sampling from **joint distribution** of t variables

- In each iteration i , there is a good choice :

- Almost fixes $\Omega(n)$ variables

- $p_{fix} \geq 1 - 1/\log^{50} n$

(bound the runtime by $n^{\text{poly}(\log n)}$)

- Bound the deviation of LP_i :

- $p_{obj} \geq 1/\log n$

- Conditioning: $E_{X \leftarrow \alpha_X}[LP_i] = LP_{i-1}$

- $ALG \leq \text{poly}(\log n) \cdot \text{opt}$

randomly **independently** sample $t = 100 \log \log n$ vars

(bound the obj by $\text{poly}(\log n) \cdot \text{opt}$)

$$(1 - p_{fix}) + (1 - p_{obj}) < 1$$

Open Questions

- **Min-3-SAT?**

- $\text{poly}(\log n)$ -approximation for complete Min-3-SAT?
- Exact Optimization is hard unless $\text{NP} \subseteq \text{BPP}$
- Quasi-poly time?
 - $n^{O(\log n)}$ is optimal for $O(1)$ -approximation Max-(2, $\text{poly}(n)$)-CSPs

- **Min-NAE-4-SAT?**

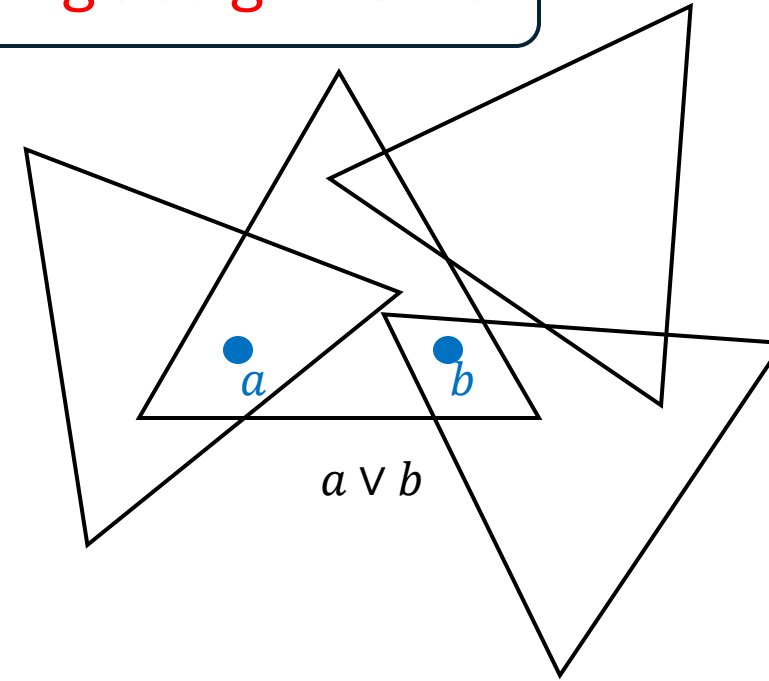
Thank You!

Toy Example: Complete 2-SAT

Thm. Any 2-SAT complete instance has $O(n)$ satisfying assignments.

Proof.

- $VC\text{-dim} \leq k \Rightarrow \#sets \leq O(n^k)$ [Sau72, She72]
- Elements: n variables
- Every satisfying assignment forms a set
 - Set of all the true variables



[Fed94] all assignments for 2-SAT can be found in time $O(n * \#satisfying\ assignments + m)$

Thm. All the satisfying assignments can be found in $O(n^2)$ time.